

Akshat Raj

AI & Generative AI Engineer | Python, LLMs, RAG Pipelines, LangChain, FastAPI, Vector Databases

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Technical Skills

AI & GenAI Frameworks: LangChain, LangGraph, AutoGen, RAG (Retrieval-Augmented Generation), CAG (Context-Augmented Generation), Prompt Engineering, LLM Integration, Agentic AI Systems, AI Orchestration (n8n)

ML & Deep Learning: Scikit-learn, Neural Networks, Deep Learning Fundamentals, Model Evaluation & Optimization, Embeddings, Semantic Search, NLP

Vector Databases & Retrieval: Vector Databases (Pinecone, FAISS, ChromaDB – conceptual & applied), Semantic Retrieval, Knowledge Retrieval Systems

Backend & APIs: Python (Pandas, NumPy, Matplotlib, Seaborn), FastAPI, REST APIs, ETL Pipelines, Data Modeling

Databases & Query: MySQL, PostgreSQL, Google BigQuery, MongoDB (Basic), SQL (CTEs, Window Functions, Aggregations)

Cloud & DevOps Tools: Google Cloud Platform (GCP), AWS (Basic), Git, GitHub, Jupyter Notebook, VS Code, Google Colab

BI & Visualization: Microsoft Power BI (DAX, Power Query), Tableau Public, Microsoft Excel (PivotTables, Macros, Power Query)

Core Competencies: Technical Documentation, System Architecture Design, Cross-functional Collaboration, AI Experimentation, Agile Methodology, Problem Solving

Experience

Data Analyst Intern, IT Department

May 2026 – Present

CESC Rajasthan, RP-Sanjiv Goenka Group

Rajasthan, India

- Designed and deployed Python-based ETL and automation pipelines to process enterprise IT datasets, enabling intelligent KPI tracking and management-ready reporting with minimal manual intervention.
- Applied data analysis and AI-assisted workflows to transform raw operational datasets into structured, decision-ready reports for senior leadership, demonstrating ability to translate business problems into data-driven solutions.
- Collaborated with cross-functional departments to gather requirements and deliver scalable analytical systems under tight enterprise deadlines.

Data Analyst Intern

Apr 2025 – Jun 2025

Thrillophilia

Remote, India

- Engineered Python (Pandas, NumPy) and SQL pipelines processing 100K+ rows of behavioral data to surface actionable insights, contributing to a **12% projected uplift in conversion rate**.
- Architected multi-page Power BI dashboards with DAX-driven KPIs tracking revenue trends and customer drop-off funnels across 5+ markets, **cutting manual reporting time by 40%**.
- Executed RFM segmentation model for personalized re-engagement campaigns, **improving repeat-booking targeting accuracy by 18%**; demonstrated applied understanding of embeddings and customer clustering logic.
- Conducted A/B test analysis with hypothesis testing (p-value, confidence intervals) that directly influenced a product UX redesign – showcasing experimentation and evaluation mindset aligned with AI model optimization.
- Automated weekly KPI reporting pipelines via Python and Excel Power Query, **saving 6+ hours per week** through end-to-end pipeline automation.

Projects

AI Competitive Intelligence Agent — LangGraph, CrewAI, ChromaDB, GPT-4o, Python, FastAPI, Redis | GitHub

2026

- Architected an autonomous multi-agent pipeline using LangGraph and CrewAI (Scraper Agent + Strategist Agent) to continuously monitor competitors via RSS feeds, Reddit API, and web sources – delivering 24/7

intelligence with zero manual intervention.

- Built a self-updating RAG system backed by ChromaDB with long-term memory, enabling semantic retrieval over a dynamically evolving knowledge base of competitor activity and market signals.
- Designed an AI orchestration layer with BullMQ job queues and cron scheduling for automated agent runs; integrated Slack Bolt SDK to push real-time alerts and summaries directly to stakeholder channels.
- Exposed system insights via FastAPI REST endpoints and documented full system architecture, agent workflows, and data flow diagrams for cross-functional handoff.

Retail Analytics Pipeline — Python, Pandas, Matplotlib, openpyxl | GitHub

Feb 2026

- Engineered a scalable data pipeline processing **32.4M+ records** across 3 heterogeneous sources; reduced memory footprint by **65%** through dtype-level optimization with full Git version control.
- Computed 13 business KPIs including INR 818 Cr revenue and 32.44% gross margin; identified 2 departments driving **44% of total revenue**, enabling data-backed category prioritization.
- Implemented RFM scoring across 206,209 customers; Champions cohort showed **10x higher spend** vs. Lost segment, with 24,123 at-risk customers flagged as the highest-ROI re-engagement target.

Customer Shopping Behavior Analysis — Python, MySQL, FastAPI, Power BI, DAX

May 2026

- Architected an end-to-end analytics pipeline across 3,900 records: cleaned raw CSV data in Python, loaded a fully normalized relational schema into MySQL, and exposed insights via a FastAPI REST API layer.
- Authored 12 SQL business queries using CTEs, window functions, and GROUP BY aggregations to surface revenue by category, repeat-purchase rates, and subscription spend differentials.
- Developed a multi-page Power BI dashboard that replaced 2 static Excel reports and **cut ad-hoc query turnaround from hours to under 5 minutes**, adopted as the team's self-serve reporting tool.

Education

VIT Bhopal University

Aug 2023 – May 2027

B.Tech, Computer Science and Engineering – Artificial Intelligence and Machine Learning | CGPA: 8.63/10 Bhopal, MP

Relevant Coursework: Deep Learning, Neural Networks, Natural Language Processing, Machine Learning, Data Structures and Algorithms, Database Management Systems, Python Programming, Cloud Computing

G.S. Janghid School, CBSE

2022 – 2023

Senior Secondary, Class XII: 83%

Jodhpur, Rajasthan

St. Anne's School, CBSE

2020 – 2021

Secondary, Class X: 90%

Jodhpur, Rajasthan

Certifications

- Applied Machine Learning in Python, Coursera, University of Michigan – Scikit-learn, Neural Networks, Model Evaluation 2025
- Cloud Computing, NPTEL, IIT Kharagpur – Cloud Architecture, GCP, Distributed Systems 2025
- Google Data Analytics Professional Certificate, Coursera (Google) – Data Analysis, SQL, Python, Tableau *In Progress*
- Machine Learning & Data Science Bootcamp, Udemy (Krish Naik) – ML Algorithms, Python, Deep Learning, NLP 2025